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Phoma spot or ascochyta spot of coffee

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The phoma stain, *Phoma* spp., is a fungal disease that occurs in several countries around the world where coffee is grown in areas of higher altitudes. The aerial part of the coffee plant can harbor several species of the genus *Phoma*. The gender *Phoma* comprises more than 2.000 species that are grouped into nine sections, and some species are related as important pathogens of the aerial part of the coffee plant, constituting one of its most important fungal diseases, as it occurs from the formation of seedlings in the nursery to the coffee production phase.

In Brazil, the etiological agents of phoma spot are fungi *Phoma tarda* (RB Stewart) H. Verm. and *Phoma costarricensis* Echandi, 1957 (Pleosporales: Didymellaceae). THE *P. late* was first observed in Ethiopia in 1954 and the *P. costarricensis* was first described in 1957 in Costa Rica. Both are responsible for one of the most important coffee diseases in Brazil, phoma spot or ascochyta spot.

The phoma stain, *P. late*, which is synonymous with *Ascochyta tarda* Stewart 1957 and *Ascochyta coffeae* Hennings 1902, and is the main etiological agent of tip blight and phoma leaf spot on coffee plants in Brazil and Africa.

In some coffee-producing regions, the disease has gained importance due to the occurrence of lesions on leaves and branches and the fall of fruits, compromising fruiting and future crop production.

Disease damage

The disease causes damage such as defoliation, loss of flower buds due to necrosis in the branches or rosettes, which can result in the indirect fall of flowers, in addition to mummification and loss of the shoots, dry rot of tips and ends of branches, resulting in losses of between 15 and 43% of the coffee tree's production.

In the field, the infection begins at the apical part, in the terminal shoot, in the tender leaves, which still have no physical barriers of resistance, reaching the young tissues. When the lesions reach the edges of the leaves, they curve, may present cracks and the symptom can easily be confused with boron deficiency (Fig. 1).

On the leaves of the second and subsequent pairs of leaves, the disease produces circular spots of dark color and varying sizes, which can reach 2 cm in diameter (Fig. 2).

In the affected branches, a symptom also called dry shoots (Fig. 3), dark depressed lesions are seen, which can involve the entire diameter.

Phoma spot can also affect floral rosettes, indirectly causing necrosis of flowers and fruits (Fig. 4).

At temperatures below 20°C, relative humidity below 70% and strong winds, the damage is more severe, especially in regions with altitudes above 900 m, even in the absence of rain, but with sufficient humidity to cause at least four hours of leaf wetness. The penetration of the fungus is facilitated by mechanical damage caused by insects to the plant tissues, or due to the collision between the tender leaves caused by winds in situations of low relative humidity below 50%. At temperatures ranging from 15 to 20°C there is a significant increase in the length of the fungus' germ tube, favoring infection by the pathogen and culminating in the progression of the disease with the increase in periods of leaf wetness.

Regardless of the type of *Phoma* spp. involved, the fungus attacks leaves, flowers, fruits and branches of the coffee plant, producing very characteristic lesions. The damage caused has a direct impact on production, since new shoots and flower buds die, fruits fall and the fruit sets poorly due to defoliation, compromising the development and future production of the plant.

The indirect damage caused by the disease is defoliation and drying of the ends of the branches, which leads to a reduction in production for the year and the following year, in addition to producing a lower quality drink.

Successive attacks of the disease promote intense sprouting of lateral branches, reducing aeration and light penetration into the interior of the coffee plant. Sporadic outbreaks can also occur in seedlings, in nurseries or in the field, if the plants are exposed to cold winds, even at altitudes below 900 m.

The phoma leaf spot has been increasing in economic importance in recent years due to the losses it causes to production. It has been a major problem for coffee plantations in high altitude regions or on hilltops, as is the case in the state of Espírito Santo and some areas in the Zona da Mata region of the state of Minas Gerais. The greatest losses are seen in plantations subject to the action of cold winds, especially in years with excessive winter rainfall. The disease can limit coffee planting in high regions, such as the plateaus of the Triângulo Mineiro, southern Minas Gerais and western Bahia.

Phoma spot therefore represents a potentially serious threat to coffee crops, but climate change may reduce the prevalence of environmental conditions favorable to its spread.

Phoma leaf spot control

Control of phoma leaf spot should begin with the adoption of a series of cultural measures that aim to prevent the onset of the disease and facilitate chemical control, if necessary. It is known that, in regions with a climate that is very favorable to the disease, the use of chemical fungicides alone does not provide satisfactory results in controlling the disease.

• **Cultural control:** It is recommended to avoid planting the crop in places where the disease is likely to develop. In regions with constant winds, it is advisable to install windbreaks. Nurseries should have good drainage and be protected from cold winds.

Mineral nutrition, or fertilization, when applied in an appropriate and balanced manner, in addition to positively affecting agricultural productivity, directly influences plant health, providing greater resistance to biotic and abiotic stresses. Among the biotic factors, the increase in plant resistance to diseases stands out, protecting them from new infections and reducing the intensity of existing infections. On the other hand, both excess and shortage of some nutrients can favor diseases, making plants more predisposed to infection. Thus, mineral nutrients can increase or reduce the intensity of plant diseases, often determining the resistance or susceptibility of the host.

• **Chemical control:** In areas where conditions are favorable for the development of phoma leaf spot, chemical control, with foliar sprays, should be carried out with fungicides (Table 1) as a preventative measure, starting in the fall-winter, targeting pre-

flowering and post-flowering. Control should continue until all coffee fruits reach the lead stage, since the coffee tree has several flowering periods.

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Quadro 1 - alguns produtos indicados e registrados para uso no controle da mancha de phoma, *Phoma* spp., em cafeeiro no Brasil

Nome		Dose p.c./ha	Formulação	Grupo Químico
Comercial	Ingrediente ativo			
Álibi	Azoxistrobina + Difeconazol	300-400 mL	SC	Estrobilurina e Triazol
Amistar WG	Azoxistrobina	100 g	WG	Estrobilurina
Aproach Power	Picoxistrobina + Ciproconazol	800-1000 mL	EC	Estrobilurina e Triazol
Aumenax	Fluxapiroxade + Oxicloreto de cobre	800-1200 mL	SC	Carboxamida e Inorgânico
Azimut Custodia	Azoxistrobina + Tebuconazol	750 mL	SC	Estrobilurina e Triazol
Bioprogress	Hidróxido de cobre	3000-5000 g	WP	Inorgânico
Bravonil 750 WP	Clorotalonil	2000-5000 g	WP	Isoftalonitrila
Cantus	Boscalida	150 g	WG	Anilida
Caramba 90	Metconazol	1500-2000 mL	SL	Triazol
Cercobin 875 WG	Tiofanato-Metílico	500-700 g	WG	Benzimidazol
Comet	Piraclostrobina	500-700 mL	EC	Estrobilurina
Curygen EC	Difeconazol	150-200 mL	EC	Triazol
Folicur 200 EC	Tebuconazol	1000 mL	EC	Triazol
Helmstar Plus	Tebuconazol + Azoxistrobina	750-1000 mL	SC	Triazol e Estrobilurina
Miravis Duo	Difenoconazol + Pidiflumetofem	300-600 mL	SC	Triazol e Pirazol Carboxamida
Nativo	Trifloxistrobina + Tebuconazol	750-1000 mL	SC	Estrobilurina e Triazol
Orkestra SC	Fluxapiroxade + Piraclostrobina	300-500 mL	SC	Carboxamida e Estrobilurina
Ozean	Azoxistrobina + Difeconazol	300-400 mL	SC	Estrobilurina e Triazol
Priori Top	Azoxistrobina + Difeconazol	300-400 mL	SC	Estrobilurina e Triazol
Redshield 750	Óxido Cuproso	1800-3000 g	WP	Inorgânico
Rovral SC	Iprodiona	1000 mL	SC	Dicarboximida
Sonora	Metominostrobina + Tebuconazol	580-725 mL	EC	Estrobilurina e Triazol
Teburaz	Azoxistrobina + Tebuconazol	750 mL	SC	Estrobilurina e Triazol
Tridium	Azoxistrobina + Mancozebe + Tebuconazol	2400-2800 g	WG	Estrobilurina, Alquilenobis e Triazol
Unizeb Glory	Azoxistrobina + Mancozebe	2500-3000 g	WG	Estrobilurina e Alquilenobis

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